Non-Profit Quiz/Tutor Bot

Course Name: GenAI

Institution Name: Medicaps University – Datagami Skill Based Course

Student Name(s) & Enrolment Number(s):

SR NO	STUDENT NAME	ENROLMENT NUMBER
1	KRATIKA SINGHAL	EN22CS303029
2	SUJOY CHAKRAVARTY	EN22CS304063
3	SAMARTH RAO	EN22CS303049
4	UJJAWAL SISODIYA	EN22CS303057
5	PARTH REVARAM	EN22CS304044

Group Name:09D11

Project Number:GAI35

Industry Mentor Name: AASHRUTI SHAH

University Mentor Name: DR. HEMLATA PATEL

Academic Year: 2022-2026

Problem Statement & Objectives

1 Project Objectives
2 Scope of the Project

Proposed Solution

1. Key features *(Just mention key features here no need to go into details)*
2. Overall Architecture / Workflow
3. Tools & Technologies Used (*If applicable)*

Results & Output

Add the below details here:

1. *Screenshots / outputs*
2. *Reports / dashboards / models*
3. *Key outcomes*

Conclusion

Mention a brief conclusion about your project summing up everything you have worked on and the key learning.

Future Scope & Enhancements

1. Problem Statement and Objectives

1.1 Problem Statement

Non-Profit organizations often struggle with training individuals in donor communication. Traditional learning methods lack personalization and real-time feedback. There is a need for an intelligent system that can:

- Assess user understanding
- Provide detailed feedback
- Improve learning outcomes interactively

Most existing systems provide only static learning materials, which do not adapt to the user's level of understanding. This leads to ineffective learning and poor engagement.
Therefore, there is a need for an intelligent system that can:

- Understand user responses
- Evaluate them contextually
- Provide personalized feedback
- Improve learning outcomes dynamically

1.2 Project Objectives

The main objectives of this project are:

- To design and develop an AI-powered chatbot for Non-Profit education
- To analyze donor email responses using NLP techniques
- To evaluate user input using a Large Language Model (LLM)
- To generate meaningful feedback rather than simple right/wrong outputs
- To identify user mistakes and explain them clearly
- To provide suggestions for improvement
- To create a continuous learning loop for users
- To make the system scalable and adaptable for real-world use

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1.3 Scope of the Project

The scope of the project includes:

- Focus on **donor email communication training**
- Implementation of **interactive assessments**
- Use of **AI models for evaluation and feedback**
- Development of a **chat-based user interface** The project can be extended to:
- Fundraising strategy training
- Volunteer communication training
- Proposal writing assistance
- NGO management education

However, the current system is limited to educational purposes and does not directly interact with real donors.

2. Proposed Solution

High-Level Workflow (Detailed Explanation)

1. The system presents a **donor email scenario** to the user
2. The user submits a response
3. The system preprocesses the input using NLP techniques
4. The response is converted into **vector embeddings**
5. The embeddings are stored/retrieved from a **Vector Database** 6. The processed input is sent to the **LLM API** 7. The LLM evaluates:
   - Relevance
   - Tone
   - Clarity
   - Persuasiveness
8. The system generates:
   - Score
   - analysis
   - explanation
   - Mistake
   - Correct answer
9. Feedback is displayed to the user
10. The user improves and retries → **Learning Loop**

2.1 Key Features

- **Interactive Chatbot Interface**
  Allows users to communicate naturally with the system.
  Scenario-Based Learning
  Users are given real-world donor email situations.
- **AI-Powered Evaluation**
  Responses are evaluated using LLMs instead of rule-based logic.
- **Contextual Feedback**
  The system explains *why* an answer is incorrect.
  Personalized Learning Experience
  Feedback adapts to the user's response quality.
- **Vector-Based Knowledge Retrieval**
  Stores embeddings for efficient similarity search.

Continuous Learning Loop
Users improve iteratively through repeated interaction.

- **Scalable Architecture**
  Can handle multiple users and large datasets.

2.2 Overall Architecture/ Workflow

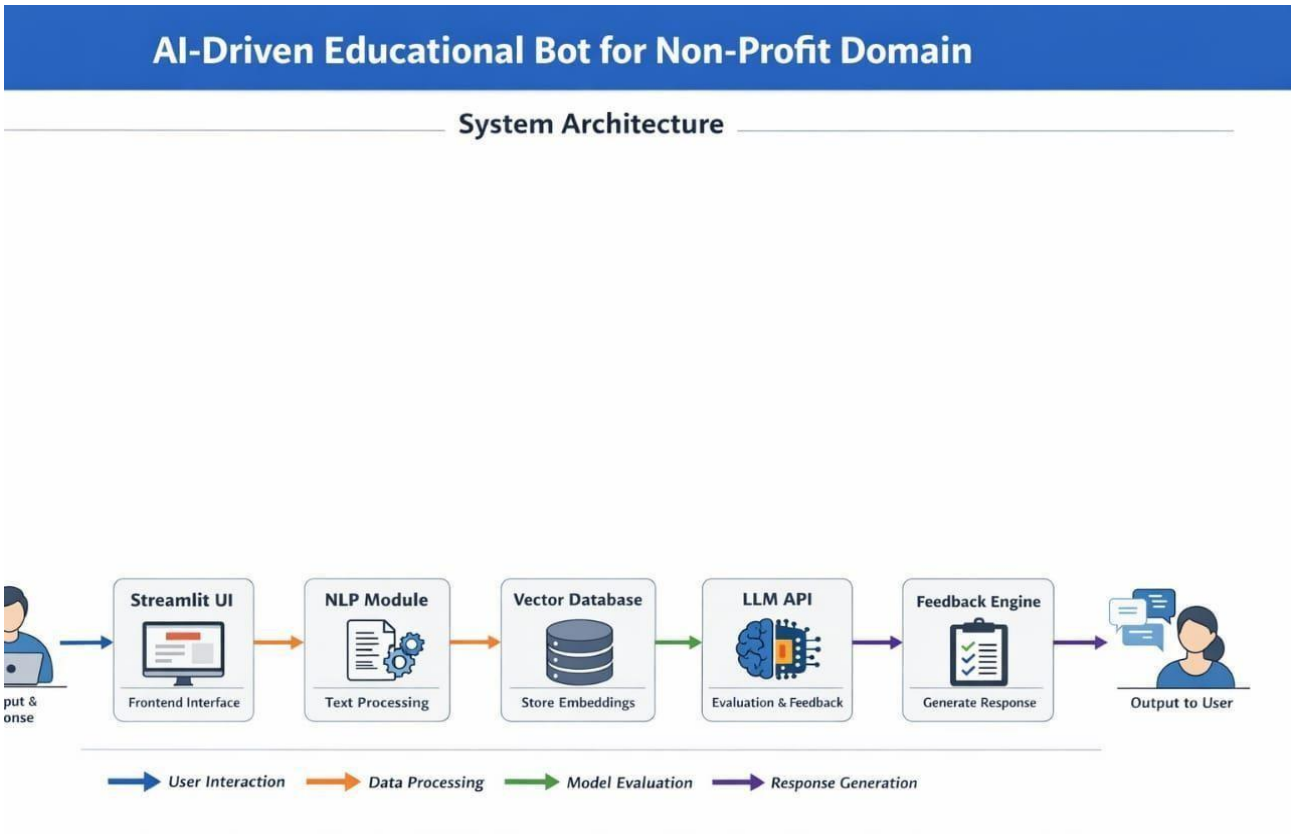
Components Explanation

1. **Frontend (Streamlit UI)** ○ Provides user interface
   - Accepts input and displays output
2. **Backend Server** ○ Handles processing logic ○ Connects all components
3. **NLP Module**
   - Cleans and preprocesses text
   - Performs tokenization, entity recognition
4. **Vector Database** ○ Stores embeddings of donor emails ○ Enables semantic search
5. **LLM API**
   - Core intelligence of the system
   - Evaluates responses and generates feedback
6. **Feedback Engine** ○ Formats LLM output ○ Generates structured explanations

2.3 Architecture Diagram:

The system follows a pipeline architecture:
User → UI → Backend → NLP → Vector DB → LLM → Feedback → User
This ensures:

- Modularity
- Scalability
- Efficient data flow



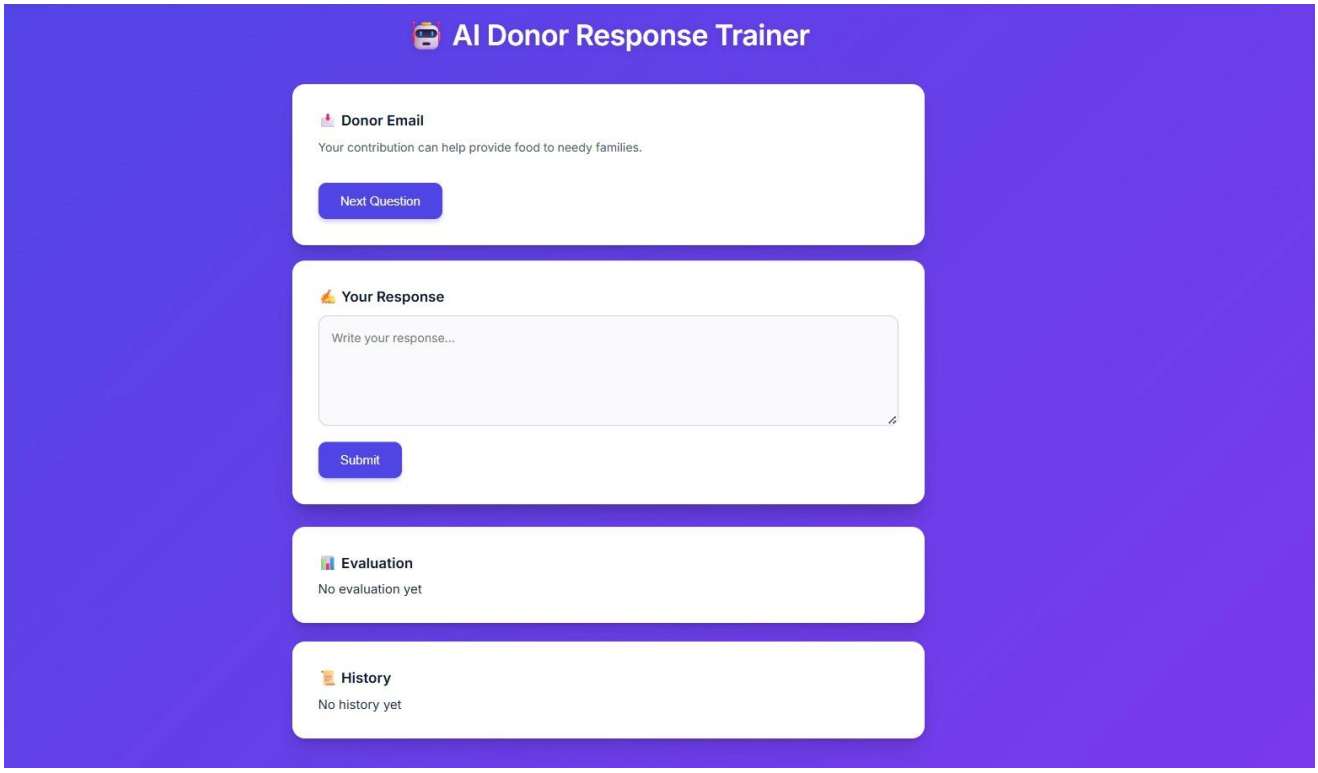
2.4 Tools and Technologies

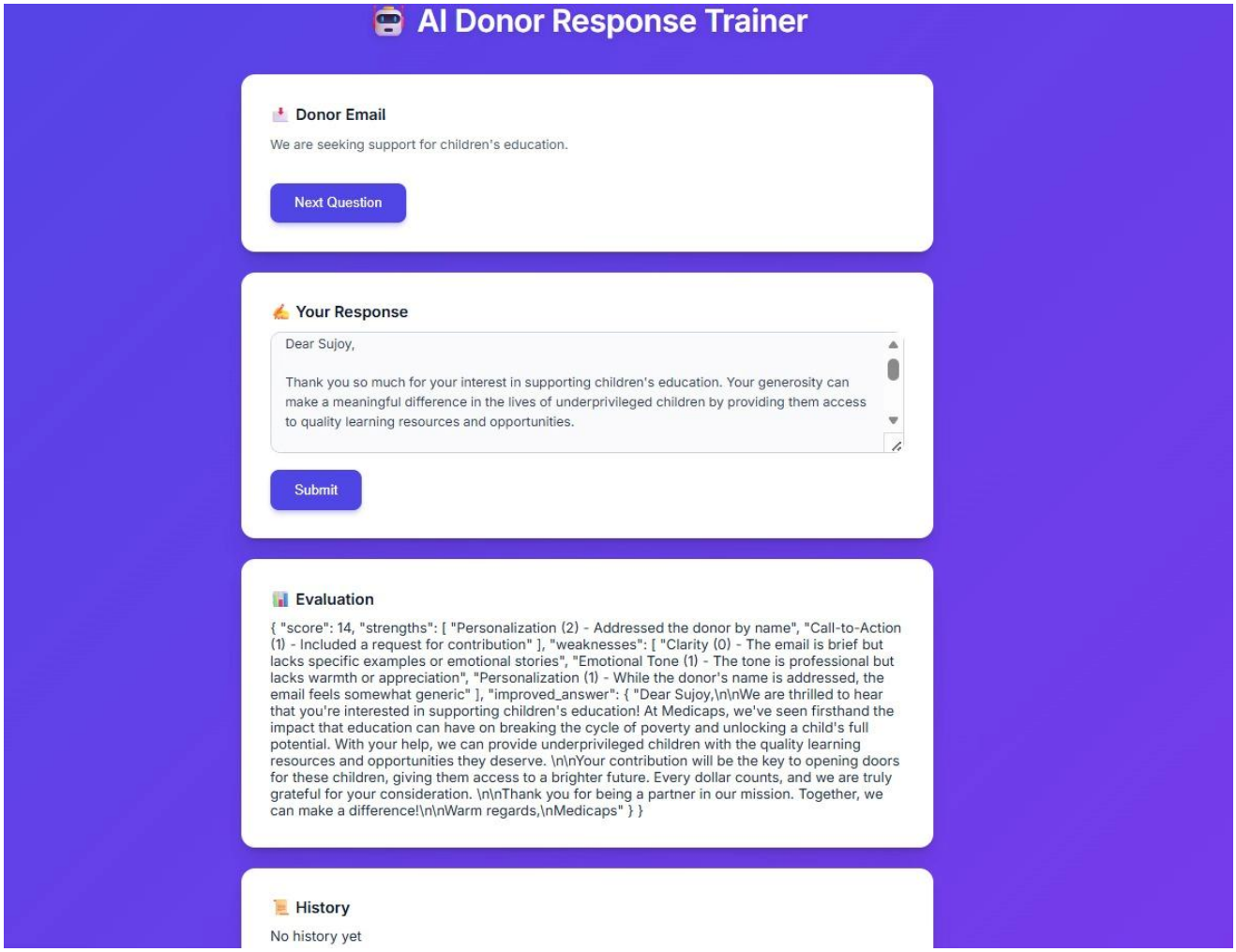
Tool	Purpose
Python	Core programming language
Streamlit	Frontend UI development
spaCy	NLP processing
Regex	Pattern matching
Scikit-learn	Basic ML tasks
OpenAI / HuggingFace	LLM-based evaluation
FAISS / Pinecone	Vector database
MongoDB	Data storage
Matplotlib	Visualization

4 Results and Outputs

The system produces:

- Accurate scoring of responses
- Detailed explanations for mistakes
- Suggested improvements
- Improved user learning over time Example Output:
- Score: 7/10
- Mistake: Lack of emotional appeal
- Suggestion: Add gratitude and impact statement




AI Donor Response Trainer
Donor Email
We are seeking support for children's education.
Next Question
Your Response
Dear Sujoy,
Thank you so much for your interest in supporting children's education. Your generosity can make a meaningful difference in the lives of underprivileged children by providing them access to quality learning resources and opportunities.
Submit
Evaluation
{ "score": 14, "strengths": ["Personalization (2) - Addressed the donor by name", "Call-to-Action (1) - Included a request for contribution"], "weaknesses": ["Clarity (0) - The email is brief but lacks specific examples or emotional stories", "Emotional Tone (1) - The tone is professional but lacks warmth or appreciation", "Personalization (1) - While the donor's name is addressed, the email feels somewhat generic"], "improved_answer": { "Dear Sujoy,\n\nWe are thrilled to hear that you're interested in supporting children's education! At Medicaps, we've seen firsthand the impact that education can have on breaking the cycle of poverty and unlocking a child's full potential. With your help, we can provide underprivileged children with the quality learning resources and opportunities they deserve. \n\nYour contribution will be the key to opening doors for these children, giving them access to a brighter future. Every dollar counts, and we are truly grateful for your consideration. \n\nThank you for being a partner in our mission. Together, we can make a difference!\n\nWarm regards,\nMedicaps" } }
History
No history yet

5 Conclusion

- The project successfully demonstrates how AI can enhance education in the Non-Profit domain. By combining NLP, LLMs, and vector databases, the system provides an intelligent, scalable, and interactive learning platform.
- It transforms traditional learning into a **dynamic, feedback-driven experience**, making it highly effective.

6 Future Scope & Enhancements

Voice-based chatbot integration
Multi- language support
Advanced emotional analysis
Integration with real NGO systems
Mobile application deployment
Use of advanced Transformer models